

Name: JOVELYN B. LOMPOT

IT 308 – ONLINE PERFORMANCE TASK

“Innovating the Future: Emerging Technology Application Design”

Phase 1: Technology Selection

Chosen Technology: Artificial Intelligence

Proposed App Title: AgriSense AI

Phase 2: Application Concept

App Name

AgriSense AI: Smart Farming Assistant for Southern Leyte

Problem Statement

Farmers in Southern Leyte often experience crop losses due to unpredictable weather, plant diseases, and lack of real-time agricultural guidance. Many small-scale farmers rely on traditional farming practices without access to modern technology or expert consultation. This results in low productivity and financial instability.

There is a need for an affordable, intelligent farming assistant that can help farmers make data-driven decisions to improve crop yield and reduce risks.

Technology Explanation

Artificial Intelligence (AI) is a technology that enables machines to simulate human intelligence. It allows systems to learn from data, recognize patterns, and make decisions with minimal human intervention.

How does it work?

AI works using:

- Data collection
- Algorithms

- Pattern recognition
- Predictive analysis

For AgriSense AI:

1. Farmers input crop data or upload plant images.
2. AI analyzes the information.
3. The system predicts diseases, weather risks, or fertilizer needs.
4. The app provides recommendations.

Application Features

1. **Crop Disease Detection**
Farmers can upload plant photos, and AI identifies possible diseases.
2. **Weather-Based Farming Alerts**
The system provides localized weather forecasts and planting advice.
3. **Smart Fertilizer Recommendation**
AI suggests the correct type and amount of fertilizer.
4. **Yield Prediction Dashboard**
Predicts harvest output based on soil and weather data.
5. **AI Chat Assistant (Bilingual: English & Cebuano)**
Farmers can ask farming-related questions anytime.

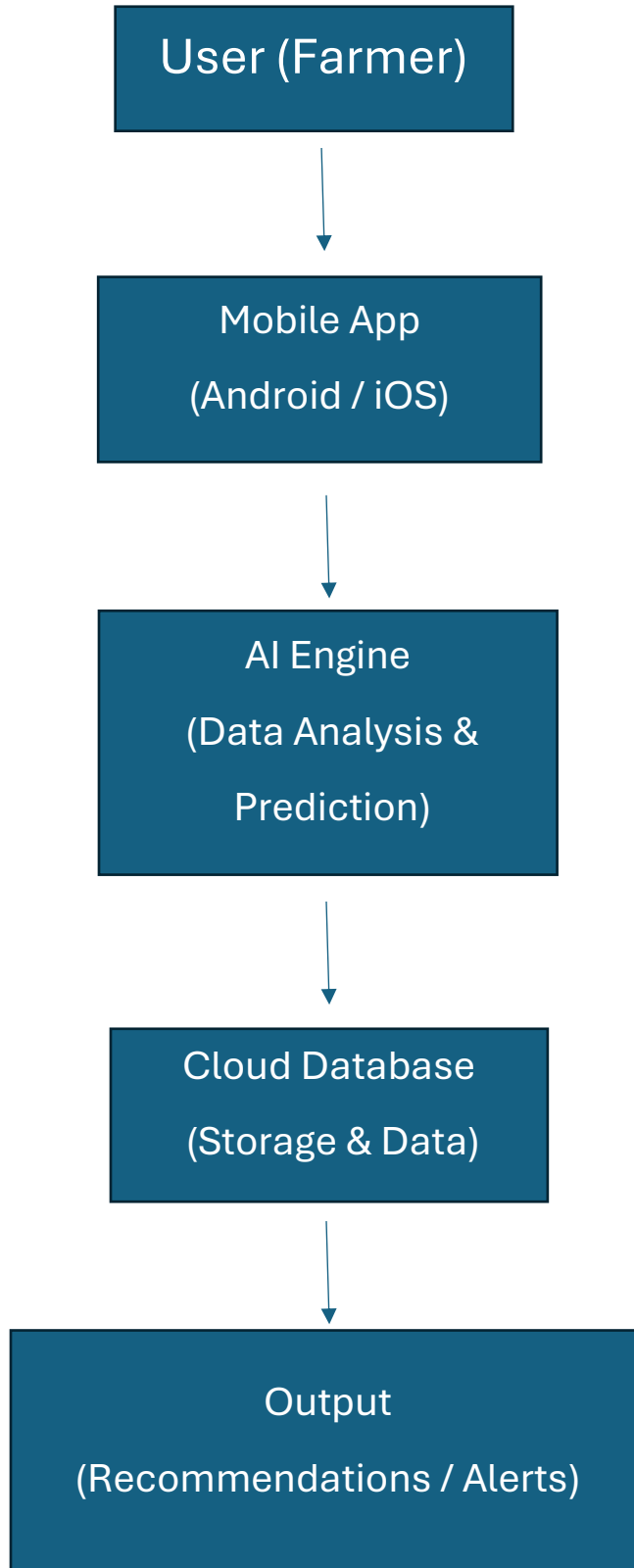
Target Users

- Farmers in Southern Leyte
- Agricultural students
- Local agricultural offices (LGUs)
- Farming cooperatives

Community Impact

- AgriSense AI can significantly improve agricultural productivity in Southern Leyte. By giving farmers access to smart recommendations, crop losses can be reduced. Increased harvest means higher income and improved food security.
- The app also supports sustainable farming by preventing overuse of fertilizers and pesticides. This protects the environment and promotes climate-resilient agriculture in the Philippines.

Simple System Architecture Diagram



Phase 3: Enhancement Requirement

Supporting Technology: Internet of Things (IoT)

Internet of Things (IoT) refers to interconnected physical devices that collect and exchange data through the internet.

Why AI + IoT is Effective

IoT sensors placed in farms can collect:

- Soil moisture levels
- Temperature
- Humidity

This real-time data is sent to the AI engine. AI analyzes the data and provides more accurate recommendations.

AI makes decisions.

IoT provides real-time data.

Phase 4: Reflection

1. Why did you choose this technology? (Artificial Intelligence)

I chose Artificial Intelligence because it is one of the most innovative and impactful technologies in today's digital world. It has the ability to analyze large amounts of data quickly and accurately. AI can provide intelligent solutions to real-world problems. I believe it can significantly improve decision-making in different industries. In agriculture, AI can help farmers predict crop diseases and weather patterns. This is very useful for provinces like Southern Leyte where farming is a major source of income. AI systems can work continuously without getting tired. It can reduce human error in analyzing farming data. AI also supports automation, which increases productivity. Many countries are already using AI in smart farming and other industries. Learning AI will help me stay competitive in the future IT industry. I chose this technology because I see its strong potential to improve both the economy and community life in the Philippines.

2. What challenges might occur in developing this?

One major challenge in developing this application is collecting accurate and reliable data. AI systems require large datasets to function properly. Without enough agricultural data, predictions may not be accurate. Another challenge is limited internet connectivity in rural areas. Farmers in remote barangays may struggle to access the application online. The cost of IoT sensors and system infrastructure may also be expensive. Technical expertise is required to build and maintain the AI system. There may also be resistance from farmers who are not familiar with technology. Training users will require time and resources. Data privacy and security must also be considered. Weather conditions in the Philippines are unpredictable, which may affect system accuracy. Maintenance and regular system updates will also be necessary. Overall, financial, technical, and user-adoption challenges may affect the development process.

3. Is it realistic? Why or why not?

Yes, I believe this project is realistic with proper planning and support. AI technology is already being used in many industries worldwide. Smart farming solutions already exist in developed countries. This shows that the concept is achievable. In the Philippines, digital transformation is continuously growing. Government agencies are also promoting agricultural modernization. Mobile phone usage among farmers is increasing. Cloud computing services are becoming more accessible and affordable. With collaboration from agricultural experts, the system can be improved. Funding may come from government grants or private investors. The project can start small and expand gradually. While it may take time and resources, it is possible to implement in the future.

4. What skills must you improve to build this in the future?

To build this application, I need to improve my programming skills. I must learn more about Artificial Intelligence and machine learning algorithms. Understanding data analytics is also very important. I need to study mobile app development to create a user-friendly interface. Knowledge in cloud computing and database management is necessary. I should also learn how to integrate IoT devices into a system. Cybersecurity skills are important to protect user data. Problem-solving skills must be strengthened to handle technical issues. Communication skills are needed to collaborate with farmers and stakeholders. Project management skills will help in organizing development tasks. I also need to improve my research skills to gather accurate information. Continuous learning and staying updated with new technologies will help me successfully build this system in the future.